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Quantification of maternal and fetal valaciclovir exposure in a pharmacokinetic study of cytomegalovirus-infected pregnant women treated to prevent vertical transmission

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Background: In cases of maternal primary infection with cytomegalovirus (CMV-MPI) maternal treatment with oral valaciclovir 8 g/day has been shown to reduce the risk of fetal infection. The pharmacological profile of this high dosage during pregnancy is not yet known.

Objectives: To quantify maternal-fetal exposure to valaciclovir 8 g/day in a population pharmacokinetic (popPK) study.

Methods: Between October 2019 and April 2023, pregnant women referred for CMV-MPI were offered to participate following: (i) CMV-MPI <14 weeks of gestation; (ii) acceptance of valaciclovir 8 g/day; and (iii) consent for amniocentesis. Amniotic fluid was tested for (i) CMV PCR for prenatal diagnosis; and (ii) dosage of aciclovir concentration (the active form of valaciclovir). Maternal serum levels of aciclovir were also measured. Aciclovir assays in both compartments were used for popPK analysis. Pharmacokinetics were described using non-linear mixed-effect modelling.

Results: We prospectively included 119 women with their 122 fetuses. CMV-MPI occurred at a median of 3.0 (range: −12; +14) weeks of gestation. CMV-infected pregnant women were treated at a median of 12.3 (range: 4.6–21.4) weeks of gestation for a median duration of 35 days (range: 7–90 days). Median pharmacokinetic parameters (C_{min}, C_{max} and AUC_{0-24}) were all successfully defined in both maternal blood and amniotic fluid compartments. No differences in aciclovir exposure were observed between infected ($n=12$, 9.8%) and non-infected fetuses. Simulations showed that after a last maternal dose, aciclovir concentration would be undetectable in the amniotic fluid after 43–47 h.

Conclusions: In this popPK study, maternal and fetal pharmacokinetics were established using *in vivo* data. The results provide a better understanding of how this fetal therapy works.

Introduction

Human cytomegalovirus (CMV) is the most prevalent cause of congenital infection and a major cause of neurological disabilities, with a worldwide prevalence of 7 cases per 1000 live births.[1] Prenatal management of congenital CMV infection has undergone major changes over the past 5 years: (i) only maternal primary infections occurring in the pre-conception period and first trimester of pregnancy can lead to neurosensory sequelae in the infant;[2,3] and (ii) secondary prevention of fetal infection initiated as early as possible

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following the diagnosis of seroconversion using valaciclovir, at a dose of 8 g/day, reduces transplacental passage of the virus by 70%.[4–6]

The prenatal dose of 8 g/day was chosen initially as it was effective in preventing CMV disease in CMV-negative renal transplant patients receiving a CMV-positive graft.[7] Prevention of vertical transmission of the virus could be considered as a similar situation, with the fetus playing the role of host and the mother that of graft. The aciclovir concentrations that led to the clinical efficacy were not investigated in the renal transplant study. The only published pharmacological study reporting pharmacokinetic (PK) parameters after the use of valaciclovir 8 g/day was carried out in healthy male volunteers.[8] By combining the results of these two studies, we put forward the hypothesis that these reported PK exposure parameters constitute an objective to be achieved to reduce maternal viral load, prevent vertical transmission, and reduce the risk of fetal CMV infection and disease.

To date, no PK studies reporting both maternal and fetal exposure to valaciclovir, used in this indication and at these dose regimens, have been conducted to our knowledge. Results from the study conducted in healthy males should be confirmed in pregnant women before being extrapolated, as it is known that pregnancy leads to significant changes in PK and pharmacodynamics.

Our objective was to quantify maternal-fetal exposure to valaciclovir 8 g/day in a population PK (popPK) study of pregnant women infected with CMV in early pregnancy and treated with the aim to reduce the risk of fetal infection. A secondary objective was to look for any association between this exposure and the occurrence of fetal infection.

Methods

Study participants

Between October 2019 and April 2023, pregnant women referred for maternal primary CMV infection were prospectively recruited. Inclusion criteria were: (i) seroconversion occurring in the pre-conceptional period or in the first trimester; (ii) maternal acceptance of oral treatment with valaciclovir 8 g/day; (iii) maternal consent for amniocentesis to be performed; and (iv) maternal consent to participate in this research.

Ethical approval from the local committee was obtained to conduct this study (IRB #00011928).

Data collection and sampling

Maternal seroconversion was always confirmed biologically in the national reference centre for herpesviruses using serological testing (IgG, IgM, IgG avidity). Dating of maternal infection with CMV was estimated using a standardized algorithm used for all pregnant women in the cohort. Pre-conceptional period was defined from about −3 months before the conception, and the first trimester was defined from 0 to 14 weeks of amenorrhoea. The serological screening strategy at the referral centre is not described in detail here but can be found in a previous published article.[9]

Treatment with valaciclovir was offered as soon as serological diagnosis was confirmed by the national reference centre. It was recommended to continue the treatment until the amniocentesis results were known. Between October 2019 and July 2020 cases were treated with 2 g/6 h (q6h), then after July 2020 they were treated with 4 g/12 h (q12h) as in the protocol proposed in the phase 3 trial published by Shahar-Nissan *et al.*[4] However, with the q12h regimen we had observed renal acute failure cases so the previous regimen, q6h, was reused by the end of 2020 and the q12h regimen was permanently dropped from the protocols of our prenatal unit.[10] Pregnant women had a biological follow-up every fortnight with maternal serum creatinine, estimated glomerular filtration rate (eGFR) and evaluation of the liver function.

Amniocentesis was performed at least 8 weeks after the presumed date of maternal seroconversion and from 17 weeks of gestation. Two investigations were performed on amniotic fluid: (i) CMV PCR for prenatal diagnosis of fetal infection; and (ii) dosage of aciclovir concentration, with maternal serum for aciclovir concentrations. We measured aciclovir concentration as it is the final active form of valaciclovir. Aciclovir assay was performed as previously detailed, and samples from both compartments were used for the popPK analysis.[11]

Information on maternal treatment was recorded including adherence (oral declaration), type of regimen (q6h or q12h), and time elapsed between drug intake and blood or amniotic fluid sampling. Baseline maternal characteristics were also collected: age, BMI, gravidity, parity, tobacco use during pregnancy, pregnancy conception method, and type of pregnancy (singleton, multiple). All women also underwent an ultrasound examination to record estimated fetal weight, amniotic fluid volume, morphological aspect of the placenta, and presence of ultrasound features of CMV fetopathy.

PopPK analysis

Data were analysed by a population approach, using the non-linear mixed-effect modelling software program *Monolix* version 2023R1 (available at www.lixoft.eu). Parameters were estimated by computing the maximum likelihood estimator of the parameters, using the stochastic approximation expectation maximization (SAEM) algorithm combined with a Markov chain Monte Carlo (MCMC) procedure.

Several structural models for aciclovir plasma PK were investigated, i.e. one and two compartments with linear absorption and elimination. Models with first-order or zero-order absorption rate constants were also considered.

Continuous covariates (CO) were tested with the following equation, using apparent elimination clearance (CL/F), for example:

$$CL/F = \theta CL/F \cdot \left(\frac{CO}{\text{median (CO)}}\right)^{\beta CO}$$

where θCL/F is the typical value of apparent elimination clearance for a subject with the median covariate value and βCO is the estimated influential factor for the continuous covariate. Aciclovir amniotic fluid PK was modelled as an effect compartment connected to the central compartment by a first-order process with same or different input and output constants. The effect compartment was modelled as a virtual compartment of negligible volume that does not modify the plasma compartmental model.

Several error models (proportional, additive, mixed) were investigated to describe the residual variability (ϵ) in plasma and amniotic fluid. Inter-individual variabilities (IIV or η) were assumed to be exponential. The objective function value (OFV) was used to test different hypotheses regarding the structural model, the structure of the variance-covariance matrix for IIV and residual variability models. The OFV was also used to test the covariate effect(s) on PK parameter(s). A covariate was finally retained in the model if its effect was biologically plausible, if the OFV was decreased by at least 3.84 (χ^2 with 1 *df*, $P < 0.05$), and if it produced a reduction in the variability of the PK parameter (IIV).

The final model was evaluated by visual inspection of the observed versus predicted (population and individual) concentrations, and the normalized prediction distribution error (NPDE) metrics versus population-predicted concentrations and time scatter plots. The prediction-corrected visual predictive checks (pcVPC) plots were also used to evaluate the predictive performance of the model. Aciclovir plasma and amniotic fluid concentrations were simulated by the model, and the 90% CIs of the 5th, 50th

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and 95th percentiles of the simulated concentrations were overlaid to the 5th, 50th and 95th percentiles of the observed concentrations.

Statistical analysis with clinical outcomes

Individual values for the AUC_{0-24} of aciclovir concentrations in both maternal plasma and amniotic fluid were computed for each patient with Bayesian estimations from the final model. Individual predicted maximal (C_{max}) and minimal (C_{min}) concentrations were also derived. Clinical success was defined as a negative CMV PCR in amniotic fluid. Association between drug exposure parameters and clinical success was investigated using Wilcoxon tests. Statistical analysis was performed using the R software.

Dosing simulation

From the final models, Monte Carlo simulations (200 replicates) were performed in order to evaluate aciclovir exposure in maternal and amniotic fluid compartments. Different valaciclovir dose options (8 g/day) were simulated: 2 g q6h or 4 g q12h regimens.

All simulations were based on a typical pregnant woman with the covariates fixed to the median population values. The simulated PK profiles were compared with previous reported exposure parameters following 8 g/day administration in healthy non-pregnant volunteers (C_{min} and C_{max} of 9 μmol/L and 30 μmol/L, respectively).[8]

Results

Study participants

During the study period, 119 CMV-infected pregnant women with 122 fetuses (three twin pregnancies) were included (Table 1). Maternal infection with CMV occurred at a median 3.0 (range: −12; +14) weeks of gestation and, based on positive CMV PCR at amniocentesis, 12 (9.8%) fetuses were infected. Regarding data on the use of valaciclovir, CMV-infected-pregnant women were treated at a median 12.3 (range: 4.6–21.4) weeks of gestation. The median time between the dating of seroconversion (by the algorithm) and the introduction of valaciclovir was 60 days (range: 10–167). For fetuses who became infected despite being treated with valaciclovir, the median time between the estimated date of seroconversion and the date of initiation of treatment was 43 days (range: 22–134). For 24 women, treatment was started after 14.0 weeks': 5 of these presented with CMV seroconversion at the very end of the first trimester (between 11 and 14 weeks'), the others were referred to our unit by their primary care provider late after their seroconversion had been diagnosed. The valaciclovir regimen was prescribed four times or two times a day to 97 and 22 pregnant women, respectively, and the median duration of treatment was 35 days (range: 7–90 days). Three cases decided to stop treatment before the amniocentesis for personal reasons (5, 7 and 70 days before). No side effects requiring discontinuation of treatment were reported. Aciclovir concentrations were assessed in maternal blood and amniotic fluid sampled at a median 17.4 (16.7–23.6) weeks'. Ultrasound examination on the day of the amniocentesis reported normal morphological examination of the placentas and fetuses as well as normal quantification of the amniotic fluid.

Table 1. Baseline demographic and clinical data of study participants

	Pregnant women[a] ($n = 119$)
Fetuses	122 (3 twins)
Maternal age, y	33 (21–42)
BMI	21.7 (17.0–54.1)
GFR	137.5 (106.9–164.2)
Parity	
0	21 (17.6%)
≥1	98 (82.4%)
GA at seroconversion, wk	3.0 (−12; + 14)
Data on the use of valaciclovir	
GA at valaciclovir initiation, wk	12.3 (4.6–21.4)
Time between seroconversion and valaciclovir initiation, d	60 (10–167)
2 g/6 h regimen	97 (81.5%)
4 g/12 h regimen	22 (18.5%)
Duration of treatment before amniocentesis, d	35 (7–90)
Data from amniocentesis day	
GA at amniocentesis, wk	17.4 (16.7–23.6)
Placenta on US scan	All normal[b]
AF on US scan	All normal[b]
Estimation of fetal weight	All eutrophic[c]
CMV PCR^+ on amniotic fluid	12/122 (9.8%)

AF, amniotic fluid; CMV, cytomegalovirus; GA, gestational age; GFR, glomerular filtration rate (GFR estimated at a mean GA of 17.1 (7–23) wk; US, ultrasound.

[a]Data are presented as median (range), or *n* (%).

[b]Normal: placenta: thickness, morphological aspect. Amniotic fluid: estimation of the volume, morphological aspect.

[c]Eutrophic: based on Hadlock formula. For six fetuses, the estimation of fetal weight was not performed on the day of the amniocentesis.

PopPK analysis

A total of 119 and 122 samples of maternal plasma and amniotic fluid for aciclovir concentrations at steady state were available for the analysis. A one-compartment model with zero-order absorption and first-order elimination with an effect compartment linked to the central compartment satisfactorily described the aciclovir maternal plasma and amniotic fluid concentrations. For amniotic fluid, estimating different input and output rate constants for the effect compartment did not improve the fit and provided similar estimates for both constants. Therefore, a unique parameter was estimated. The PK parameters of this model were the absorption constant (Tk0), apparent elimination clearance (CL/F), volume of distribution (V/F), maternal to amniotic fluid transfer rate constant (kmp) and amniotic fluid elimination rate constant (kmp), F being the unknown bioavailability (Table 2).

The residual variability was described by a mixed error model for maternal plasma and additive model for amniotic fluid. Inter-individual variability was retained for CL/F. The most significant decrease in OFV was obtained by adding the effect of eGFR on CL/F (8 units decrease, P value = 0.0047). After the inclusion of eGFR on CL/F, the addition of gestational age (GA) on CL/F significantly decreased the OFV (4 units decrease, P value = 0.046). An additional drop in OFV was observed with BMI effect on CL/F (5 units decrease, P value = 0.025). Thus, the final covariate model was $CL/F = \theta_{CL/F} \times \left(\frac{eGFR}{137.5}\right)^{\beta_{eGFR}} \times \left(\frac{BMI}{21.7}\right)^{\beta_{BMI}} \times \left(\frac{GA}{17.4}\right)^{\beta_{GA}}$ with $\theta_{CL/F}$ the typical value of CL/F for a typical patient. Table 2 summarizes the

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Table 2. Estimation of the population parameters in the popPK analysis

Population parameter	Estimated value	RSE, %
Fixed population effects		
Absorption constant, h	2.01	1.53
Clearance, L/h	42.8	3.65
Volume of distribution, L	132	16.3
βeGFR effect on clearance	1.8	27.7
βBMI effect on clearance	0.468	46.5
βGA_at_dosage effect on clearance	−1.2	43.6
Kmp_pop	0.131	7.1
Interindividual variability		
Omega_CL	0.267	11
Residual variability		
Residual additive error (maternal concentrations)	0.22	
Residual proportional error (maternal concentrations)	0.339	9.24
Residual additive error (amniotic fluid concentrations)	5.86	12.2

AF, amniotic fluid; Beta terms, influential factors of covariates on clearance; CMT, compartment; GA, gestational age; GFR, glomerular filtration rate (GFR estimated at a mean GA of 17.1 (7–23) weeks; Kmp_pop, tranfer rate constant between maternal and amniotic fluid compartments; Omega_CL, interindividual variability on clearance; RSE, relative standard errors; US, ultrasound.

final population PK estimates including the relative standard errors. Diagnostic plots are provided in Figure S1 (available as Supplementary data at *JAC* Online).

Statistical analysis with clinical outcomes

Median (range) PK parameters were (i) in the maternal compartment: AUC_{mat} (0–24 h) = 570.2 (297.6–1282.4) µmol.h/L, $Cmin_{mat}$ = 9.6 (0.3–38.6) µmol/L and $Cmax_{mat}$ = 40.7 (27.1–87.9) µmol/L; and (ii) in the amniotic fluid compartment: AUC_{AF} (0–24 h) = 562.4 (293.6–1229.2) µmol.h/L, $Cmin_{AF}$ = 21.2 (9.3–48.8) µmol/L and $Cmax_{AF}$ = 25.4 (13.4–52.8) µmol/L.

Based on positive CMV PCR at amniocentesis, 12 (9.8%) fetuses were infected (CMV PCR^{+}). In both maternal and amniotic fluid compartments, no statistical differences were observed in aciclovir exposure between the infected and non-infected cases (Figure 1, Table S1).

Dosing simulations

The evolution of aciclovir concentration in the amniotic fluid after a maternal oral dose of valaciclovir 8 g, depending on the type of regimen (q6h versus q12h), has been simulated and the results are illustrated in Figure 2. Simulations showed that median steady-state minimal concentrations between q6h and q12h regimen were close. However, the C_{max} with the q12h regimen was 50% higher than with the q6h one, with important fluctuations. Furthermore, simulations showed that after a last maternal dose, the aciclovir concentration would be undetectable in the amniotic fluid after 43 and 47 h under q6h and q12h regimen, respectively (Figure 2). The same

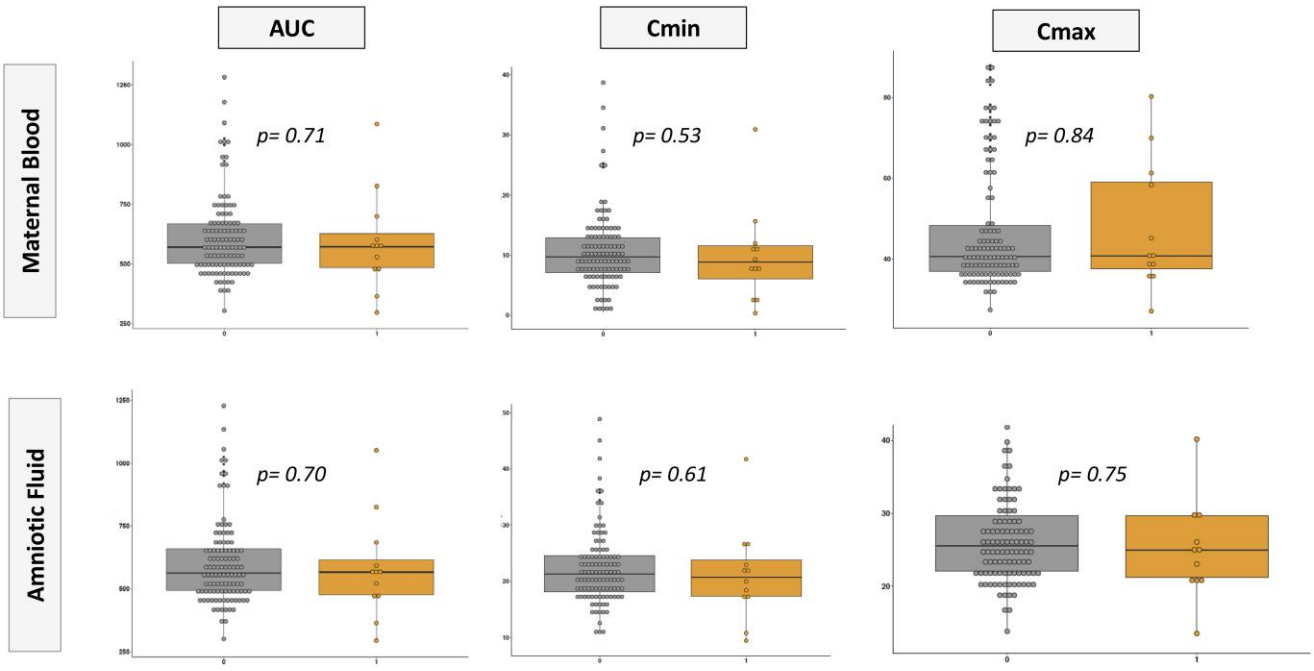


Figure 1. Comparison of median maternal and fetal pharmacokinetic parameters between pregnant women who transmitted the virus to their fetuses (1 in the X-axis) and those who did not (0 in the X-axis). The Y-axis corresponds to the value of the pharmacokinetic parameter: AUC in µmol.h/L; C_{max} and C_{min} in µmol/L. The shaded boxes correspond to the distribution of data between the 1st and 3rd interquartiles, the horizontal line in the box corresponds to the median value. AUC, area under the curve; C_{max}, maximum concentration; C_{min}, minimum concentration.

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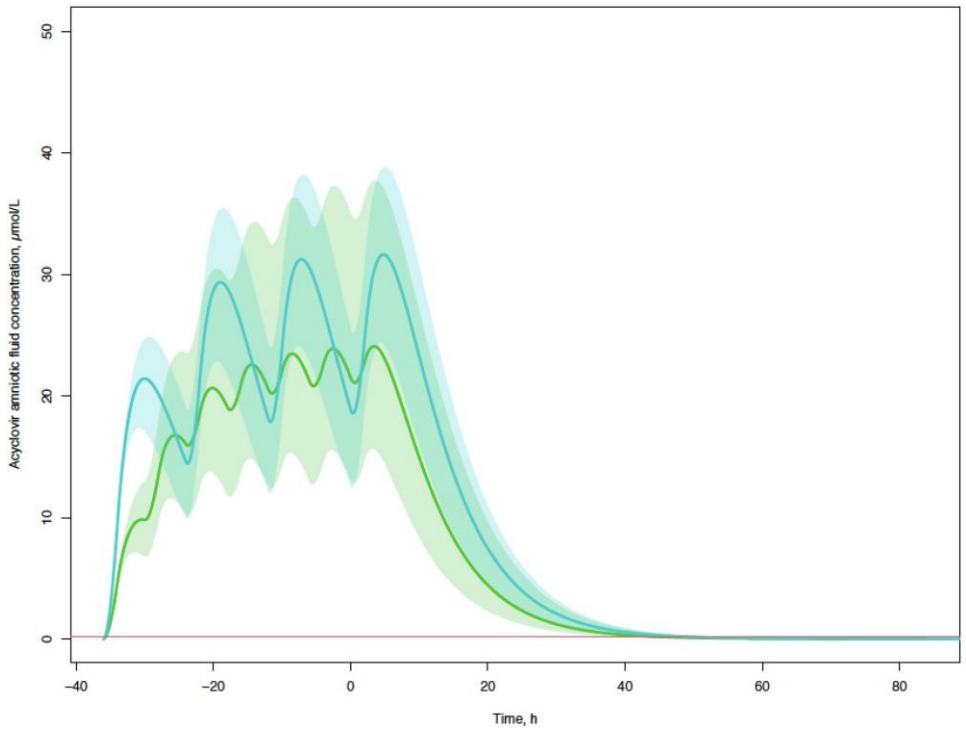


Figure 2. Aciclovir amniotic fluid concentrations (µmol/L) after maternal oral dose of valaciclovir 8 g/day: 2 g/6 h or 4 g/12 h regimens in green and blue, respectively. On the *x*-axis, $t=0$ corresponds to the last maternal oral administration of valaciclovir after steady state was reached.

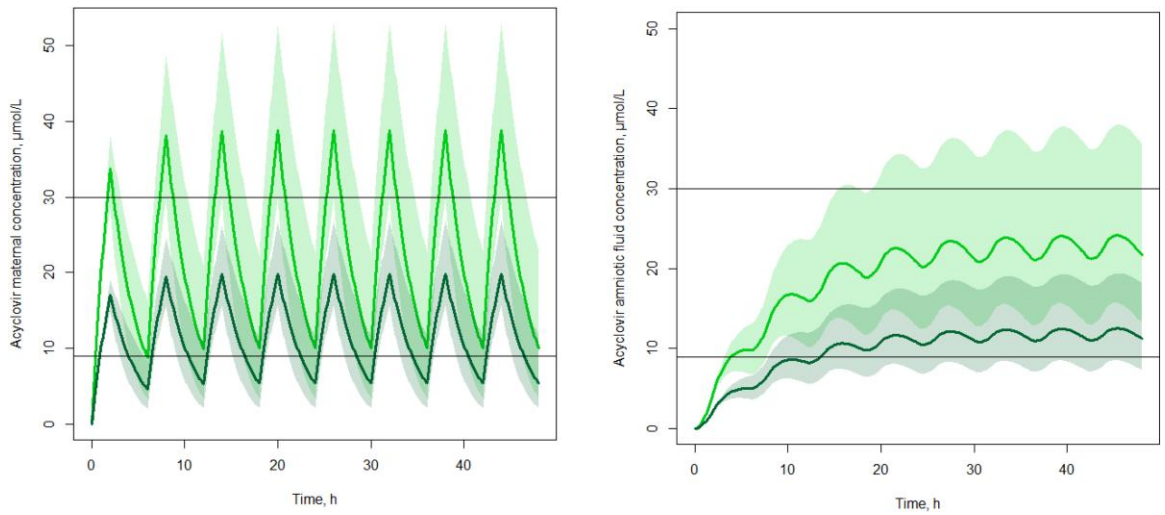


Figure 3. Median aciclovir concentrations in the maternal serum (left) and in the amniotic fluid (right), when comparing the standard dose of 8 g/day (green) versus a reduced maternal dose of 4 g/day (black). The shadded area correspond to the 5th and 95th simulated percentiles. The horizontal lines represent the reported C_{min} and C_{max} values following 8 g/day administration in healthy non-pregnant volunteers (9 µmol/L and 30 µmol/L, respectively).

simulation has been performed in the maternal blood (Figure S2). Identical fluctuations in concentrations and the same variation in C_{max} (50% difference) were also observed in the mother. Valaciclovir was undetectable in maternal blood approximately 18 h after the last dose.

Simulations were also performed to compare the standard dose of 8 g/day versus a reduced maternal dose of 4 g/day (1 g q6h) (Figure 3). Simulated maternal plasma concentrations using 8 g/day were within the range of reported values in healthy non-pregnant volunteers.[8] However, the 4 g/day dose provided maternal minimal aciclovir concentrations below 9 µmol/L in more than 50% of women. Regarding the aciclovir exposure in amniotic fluid, the standard dose of 8 g/day provided concentrations above 9 µmol/L in more than 99% of pregnancies. When

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using the half dose, there were more than 75% of pregnancies with amniotic fluid concentrations remaining above 9 μmol/L.

Discussion

Principal findings

The popPK analysis, carried out in pregnant women infected and treated with 8 g/day valaciclovir, showed that the concentrations measured in the maternal and amniotic fluid compartments were within the range of therapeutic levels previously reported as being able to control CMV infection.[7,8] The PK of aciclovir in pregnant women, when compared with that in healthy volunteers adults, was not substantially modified by either maternal-fetal CMV infection or gestational age in the first and second trimesters. Exposure PK parameters were identical whether vertical transmission of the virus had occurred or not, suggesting that transplacental passage of valaciclovir was not altered by CMV infection. Therefore, in this cohort, treatment failures (i.e. when vertical transmission occurred), do not appear to be related to drug underexposure.

We also showed that aciclovir in amniotic fluid remained detectable only 48 h after maternal last intake of valaciclovir. This information is important to enable patients to optimize adherence to treatment.

Results in the context of what is known

To date, the main hypothesis to explain valaciclovir failure is the timing of treatment initiation. Indeed, it has been reported that (i) a short interval between maternal seroconversion and the introduction of treatment,[4] and (ii) an early gestational age at the initiation of valaciclovir[5] were major predictive factors of success.

In terms of toxicity, it has been shown that when the maternal dosage of 8 g/day was given in two doses, maternal acute obstructive renal failure was observed, whereas no such complications were observed when the dose was split into four doses.[10] The pathogenesis of valaciclovir-related acute renal failure mainly involves aciclovir accumulation and precipitation of crystals in renal proximal tubule cells.[12] In this study, we analysed the variations in concentrations in the amniotic compartment with the two dosing regimens: the C_{max} with the q12h regimen was 50% higher than with the q6h one, with important fluctuations suggesting that fetal toxicity might also be expected with this q12h scheme. These data reinforce our recommendation to treat four times a day.

Clinical implications

Our study also showed that, even if the fetus is infected, the estimated concentrations in the amniotic compartment should theoretically be sufficient to prevent fetal CMV disease to happen. Indeed, the concentrations observed in our study are within the same ranges as those probably observed in CMV-negative kidney transplant recipients receiving a CMV-positive graft and in whom a reduction in CMV disease has been observed.[7,8]

These data are important from a clinical standpoint, since continuing valaciclovir after a positive amniocentesis may be beneficial to contain viral replication and limit the severity of lesions initiated at an earlier stage of embryonic and fetal development.[13,14] Evidence for the latter is limited to a Phase 2 study in infected fetuses showing signs of infection on ultrasound scans in the second trimester of pregnancy following maternal primary infection before 2016, therefore without valaciclovir secondary prevention.[15] In addition, a reduction in the rate of terminations of pregnancy for severe forms of fetal CMV disease was observed following failure of valaciclovir secondary prevention in comparison with infected fetuses in the absence of valaciclovir preventive treatment.[5] However, larger studies describing the long-term outcome of infected fetuses resulting from failure of valaciclovir prophylaxis 8 g/day in early pregnancy are awaited.

Research implications

The dose of 8 g/day has been shown to be effective in preventing vertical transmission, but this dose has not yet been challenged. This high dosage is cumbersome for pregnant women, representing 8 to 16 tablets per day depending on the dose per tablet, and is associated with a risk of renal toxicity. In our study, we showed that 8 g/day provided amniotic fluid concentrations far above the reported C_{min} values in healthy volunteers. When the dose was halved, we found that aciclovir concentrations in the amniotic fluid remained above the reported C_{min} in more than three-quarters of pregnancies. Our study provides pharmacological support for the design of clinical studies aiming at reducing doses.

Strengths and limitation

Evaluation of the PK parameters of valaciclovir 8 g/day used to treat CMV-infected pregnant women to prevent vertical transmission of the virus has long been missing; our study described this in a large cohort of 119 infected pregnant women and their 122 fetuses.

The main limitation of our study is that aciclovir assays were carried out in the amniotic fluid and not in fetal blood. The results obtained are therefore only an indirect reflection of actual fetal exposure. However, aciclovir concentrations from fetal blood have been reported in a single previous study including several severely affected CMV-infected fetuses.[16] Fetal blood concentrations were, in the 10 fetuses studied, lower than those measured in the amniotic compartment. There is currently no tool to correlate concentrations observed in amniotic fluid and in fetal plasma. Compartmental modelling studies could help to extrapolate these results.[17]

Conclusion

In this popPK study assessing the use of valaciclovir 8 g/day during pregnancy, maternal and fetal PK parameters were established using *in vivo* data. The results obtained provide a better understanding of how this fetal therapy works. It should help in refining the optimal therapeutic management of maternal seroconversions and fetal CMV infections during pregnancy.

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Transparency declarations

In the context of this work, none of the authors has any financial conflicts of interest.

Supplementary data

Figures S1 and S2 and Table S1 are available as Supplementary data at *JAC* Online.

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